

Section - B

Q. 2(i)(or)

A: 2(i)(or)

Inclination: θ

$$\theta = \tan^{-1}(m)$$

$$\theta = \tan^{-1}(-1.732)$$

$$\theta = -60^\circ$$

Inclination:

$$180^\circ - 60^\circ = 120^\circ$$

Q. 2(ii)

A: 2(ii)

Gradient:

$$A(x_1, y_1), B(x_2, y_2)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{8 - 6}{5 - 2}$$

$$= \frac{2}{3}$$

$$m = \frac{2}{3}$$

Inclination: θ

$$\theta = \tan^{-1}(m)$$

$$\theta = \tan^{-1}\left(\frac{2}{3}\right)$$

$$\theta = 33.69^\circ$$

Q: 2 (iii) (or)

A: 2 (iii) (or)

$A(3, 2)$, $B(5, 8)$
 x_1, y_1 x_2, y_2

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\Rightarrow 4 = \frac{8 - 2}{5 - 3}$$

$$\Rightarrow 4 = \frac{8 - x}{2} \quad (\text{cross multiplication})$$

$$1(8 - x) = 2 \times 4$$

$$8 - x = 8$$

$$8 - 8 = x$$

$$0 = x$$

$$x = 0$$

Q: 2 (iv)

A: 2 (iv)

$u(0, -3)$, $v(4, 7)$, $z(6, 12)$

① $m = \frac{y}{x}$

$$m = \frac{7 - (-3)}{4 - 0}$$

$$= \frac{7 + 3}{4}$$

$$m = \frac{10^5}{4_2} \Rightarrow \boxed{m = \frac{5}{2}}$$

② $m = 4_2$

$$m = \frac{12-7}{6-4}$$

$$\boxed{m = \frac{5}{2}}$$

③ $m = 1_2$

$$m = \frac{12-(-3)}{6-0}$$

$$m = \frac{12+3}{6}$$

$$m = \frac{15}{6_2} \Rightarrow \boxed{m = \frac{5}{2}}$$

Q: 2 (v)

A: 2 (v)

$(-5, 7)$

$$y - y_1 = m(x - x_1)$$

$$y - 7 = 4(x - (-5))$$

$$y - 7 = 4(x + 5)$$

$$y - 7 = 4x + 20$$

$$y - 4x = 27 + 20$$

$$y - 4x = 47$$

$$= 0$$

Q: 2 (vi)

A: 2 (vi)

$$\frac{(y-4)}{(-4-4)} = \frac{(x-2)}{(-1-2)}$$

$$\frac{(y-4)}{-8} = \frac{(x-2)}{-3}$$

$$-3(y-4) = -8(x-2)$$

$$-3y + 12 = -8x + 16$$

$$-3y + 8x = 16 - 12$$

$$-3y + 8x = 4$$

$$8x - 3y + 4 = 0$$

Q: 2 (vii) (or)

A: 2 (vii) (or)

$$4x - 3y = -9$$

$$\frac{4x}{-9} + \frac{-3y}{-9} = 1$$

$$\frac{x}{-9/4} + \frac{y}{3} = 1$$

x-intercept = $-9/4$ and y-intercept = 3

Q: 2 (vii)

A: 2 (vii)

$$y - y_1 = m(x - x_1)$$

$$y - (-2) = -\frac{1}{3}(x - 3)$$

$$\Rightarrow y + 2 = -\frac{1}{3}(x - 3)$$

$$\Rightarrow 3(y + 2) = -1(x - 3)$$

$$\Rightarrow 3y + 6 = -x + 3$$

$$\Rightarrow x + 3y + 3 = 0$$

$$\Rightarrow x + 3y + 3 = 0$$

Q: 2 (in)

A: 2 (in)

$$\tan \theta = \left(\frac{m_2 - m_1}{1 + m_1 m_2} \right)$$

$$\tan \theta = \frac{2 - (-1)}{1 + (-1)(2)}$$

$$\frac{2 - (-1)}{1 + (-1)(2)}$$

$$\tan \theta = -3$$

$$\theta = \tan^{-1}(-3)$$

$$= 71.5^\circ$$

$$= \frac{2 + 1}{1 - 2}$$

$$= \frac{3}{-1}$$

$$= -3$$

Inclination :-

$$180 - 71.5^\circ$$

$$= 108.5^\circ$$

$$\text{Q: } 2(x) \text{ ---}$$

$$\text{A: } 2(x) \text{ ---}$$

Point of intersection: $2x + y + 1 = 0$ and $x - y - 4 = 0$

$$\begin{array}{r} 2x + y + 1 = 0 \\ + \quad x - y - 4 = 0 \\ \hline 3x - 3 = 0 \end{array}$$

$$3x - 3 = 0$$

$$3x = 0 + 3$$

$$3x = 3$$

$$x = 1$$

$$\boxed{x = 1}$$

~~Solving~~ Substituting value of x in eq 2:

$$x - y - 4 = 0$$

$$1 - y - 4 = 0$$

$$-y = +4 - 1$$

$$-y = +3$$

$$y = -3$$

Point of intersection: $(1, -3)$

Q: 2 (vi)

A: 2 (vi)

Lines through intersection: ~~$6x + 5y + 3 = 0$~~ and

$x + 2y = 0$, $x - y = 3$ and point (x_1, y_1)

$$(x + 2y) + K(x - y - 3) = 0$$

$$(1 + 2(1)) + K(1 - 1 - 3) = 0$$

$$3 - 3K = 0$$

$$-3K = 0 - 3$$

$$+3K = +3$$

$$K = \frac{3}{3}$$

$$K = 1$$

$$K = 1$$

$$(x + 2y) + (x - y - 3) = 0$$

$$x + x + 2y - y - 3 = 0$$

$$2x + y - 3 = 0$$

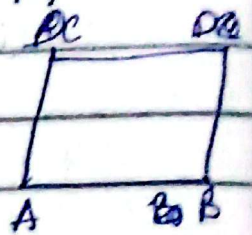
Section : C

Q: 3 (i) (or)

A: 3 (i) (or)

$A(-2, 1)$, $B(6, 3)$, $C(10, 5)$, $D(2, 3)$

slopes of AB :-



$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3 - 1}{6 - (-2)}$$

$$= \frac{2}{6+2}$$

$$= \frac{2}{8} \Rightarrow \frac{1}{4}$$

Slope of BC :-

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5 - 3}{10 - 6}$$

$$= \frac{2}{4} \Rightarrow \frac{1}{2}$$

Slope of CD :-

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3 - 5}{2 - 10}$$

$$= \frac{-2}{-8} \Rightarrow \frac{1}{4}$$

Slope of AD :-

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{3 - 1}{2 - (-2)}$$

$$= \frac{2}{4} \Rightarrow \frac{1}{2}$$

$$AB \parallel CD$$

$$BC \parallel AD$$

AB is parallel to CD
BC is parallel to AD.

Hence proved.

Q.3 (ii)

A.3 (ii)

i) Slope intercept form :-
 $6x - 5y + 15 = 0$

$$5y = 6x + 15$$

$$5y = 6x + 15$$

$$y = \frac{6}{5}x + \frac{15}{5}$$

$$y = \frac{6}{5}(x) + (3)$$

$$y = \frac{6}{5}(x) + 3$$

iv)

Two intercept form :-

(P.T.O)

formula : $\frac{x}{a} + \frac{y}{b} = 0$

$$\frac{6x}{-15} + \frac{5y}{+15} = 0$$

$$\frac{x}{-\frac{15}{6}} + \frac{y}{\frac{15}{5}} = 0$$

$$\frac{x}{-5/2} + \frac{y}{3} = 0$$

x-intercept form : $-5/2$

y-intercept form : 3

ii) Point slope :-

$$(x_1, y_1) = (0, 3)$$

$$m = \frac{6}{+5} = \frac{6}{5}$$

$$y - y_1 = m(x - x_1)$$

$$y - 3 = \frac{6}{5}(x - 0)$$

$$5(y - 3) = 6(x - 0)$$

$$5y - 15 = 6x - 0$$

$$0 = 6x - 5y + 15$$

$$6x - 5y + 15 = 0$$

ii) Two point form:

$$(x_1, y_1) = (0, 3)$$

$$(x_2, y_2) = \left(-\frac{5}{2}, 0\right)$$

~~$$y - y_1 = m(x - x_1)$$~~

~~$$y - 3 = \frac{6}{5}(x - 0)$$~~

~~$$5(y - 3) = 6(x - 0)$$~~
~~$$5y - 15 = 6x - 0$$~~

$$\frac{y - 3}{0 - 3} = \frac{x - 0}{-\frac{5}{2} - 0}$$

Gradient: m

$$\frac{y_2 - y_1}{x_2 - x_1} = m$$

$$\frac{0 - 3}{-\frac{5}{2} - 0} = m$$

$$\frac{-3}{-\frac{5}{2}} = \frac{-3 \times 2}{-5}$$

$$= \frac{6}{5}$$

iv) Normal form:

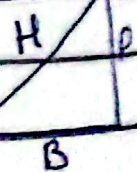
$$6x - 5y = -15$$

$$-6x + 5y = 15$$

$$\sqrt{(-6)^2 + (5)^2} = \sqrt{36 + 25} = \sqrt{61} = 15$$

$$\sqrt{36 + 25}$$

(P.T.O)



$$\sin \theta = \frac{P}{H}$$

$$\cos \theta = \frac{B}{H}$$

$$\tan \theta = \frac{P}{B}$$

$$H^2 =$$

$$\sqrt{(6)^2 + (5)^2} = 15$$

$$\sqrt{36 + 25} = 15$$

$$\sqrt{61} = 15$$

$$\frac{-6}{61}x + \frac{5}{\sqrt{61}}y = \frac{15}{\sqrt{61}}$$

* Symetric form :-

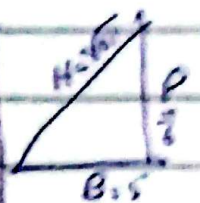
$$6x - 5y + 15 = 0$$

$$m = \tan \theta = \frac{6}{5}$$

$$\sin \theta = \frac{P}{H}$$

$$\cos \theta = \frac{B}{H}$$

$$\tan \theta = \frac{P}{B}$$



formula 1- $\frac{x-x_1}{\sin \theta} = \frac{y-y_1}{\cos \theta}$

~~$$x-0 = \frac{y-3}{\frac{5}{\sqrt{61}}}$$~~

$$\frac{x-0}{\frac{6}{\sqrt{61}}} = \frac{y-3}{\frac{5}{\sqrt{61}}}$$

$$H^2 = B^2 + P^2$$

$$H^2 = 5^2 + 6^2$$

$$H^2 = 25 + 36$$

$$\sqrt{H^2} = \sqrt{61}$$

$$H = \sqrt{61}$$

$$\begin{array}{l} \text{Q: 3 (iii)} \\ \text{A: 3 (iii)} \end{array}$$

Slopes of sides:-

① Slopes of AB:-

$$A(-2, 0) , B(3, 0)$$

$$m_{AB} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{0 - 0}{3 - (-2)}$$

$$= \frac{0}{3+2}$$

$$= \frac{0}{5} \Rightarrow 0 \times 5 = 0$$

② Slopes of BC:-

$$B(3, 0) , C(6, 5)$$

$$m_{BC} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5-0}{6-3}$$

$$m_{AC} = \frac{5}{3}$$

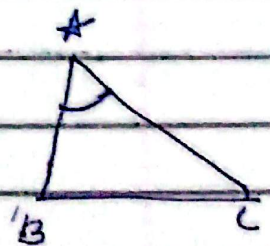
③ Slopes of $\triangle ABC$:-
 $A(-2, 0)$, $C(6, 5)$

$$m_{AC} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$= \frac{5-0}{6-(-2)}$$

$$= \frac{5}{6+2}$$

$$= \frac{5}{8}$$



Angle of A (from AB to AC)
 $\tan a = \frac{m_2 - m_1}{1 + (m_1 m_2)}$

$$\tan a = \frac{\left(\frac{5}{8} - 0\right)}{1 + (0)\left(\frac{5}{8}\right)}$$

$$= \left(\frac{5/8}{1+0} \right)$$

$$= \frac{5/8}{1}$$

$$\tan a = \frac{5}{8}$$

$$\theta = \tan^{-1}(5/8)$$

$$\theta a = 32^\circ$$

Angle of β (at from BA to BC)

$$m_{BA} = 0, m_{BC} = 5/3$$

$$\tan \beta = \frac{(m_2 - m_1)}{1 + m_1(m_2)}$$

$$= \frac{5/3 - 0}{1 + 0(5/3)}$$

$$= \frac{5/3}{1+0}$$

$$= \frac{5/3}{1}$$

$$= \frac{5}{3}$$

$$\tan \beta = \frac{5}{3}$$

$$\angle \beta = \tan^{-1} \left(\frac{5}{3} \right)$$

$$\angle \beta = 59^\circ$$

Angle at C (by subtraction):-

$$\gamma = 180^\circ - 32^\circ - 59^\circ$$

$$\gamma = 89^\circ$$

$$\text{Verification :- } 89^\circ + 32^\circ + 59^\circ = 180^\circ$$

Q: 3(iv) (or)

Ans: 3(iv) (or)

① Standard form equation:-

Let x = fruiters per dozen and y = Shakti malta per dozen.

$$120x + 150y = 1200$$

divided by 30 :-

$$\begin{array}{r} 4 \\ 120x + 150y = 1200 \\ 30 \quad 30 \quad 30 \\ \hline 4x + 5y = 40 \end{array}$$

$$4x + 5y = 40$$

(ii) 4 dozen Shakri maltas :-

$$4x + 5(4) = 40$$

$$4x + 20 = 40$$

$$4x = 40 - 20$$

$$4x = 20$$

$$x = \frac{20}{4}$$

$$x = 5$$

$$x = 5$$

(iii) Slope intercept form :-

$$4x + 5y = 40$$

formula :- $y = mx + c$

$$5y = 4x + 40$$

$$y = \frac{4}{5}x + 8$$

Slope = $-\frac{4}{5}$ means for each additional dozen
of fruiters sold, Shakri malta dozens decrease
by $\frac{4}{5}$ (fixed budget trade-off).

Total Marks: 65

General Instructions:

1. Write your name, roll number, and section clearly on the answer sheet.
2. **Section A (MCQs):** Attempt **ALL 12** questions. Each carries **1 mark**. Time: 15 minutes.
3. **Section B (Short Questions):** Attempt **ALL 11** pairs, choosing the main question **OR** its alternative. Each carries **3 marks**.
4. **Section C (Long Questions):** Attempt **ALL 4** pairs, choosing the main question **OR** its alternative. Each carries **5 marks**.
5. Use black or blue ink only. Pencil is not allowed except for diagrams.

SECTION A – Multiple Choice Questions (12 × 1 = 12 Marks)

Encircle the correct option. All questions are compulsory.

#	Question	A	B	C	D	Answer
1	The gradient of a straight line is defined as:	rise ÷ run	run ÷ rise	rise × run	rise + run	●○○○
2	The gradient of a horizontal line is:	1	undefined	0	-1	○○●○
3	If m_1 and m_2 are gradients of two perpendicular lines, then:	$m_1 = m_2$	$m_1 + m_2 = 0$	$m_1 \times m_2 = 1$	$m_1 \times m_2 = -1$	○○○●
4	Slope-intercept form of a line with slope m and y -intercept c is:	$y = mx + c$	$y - y_1 = m(x - x_1)$	$x/a + y/b = 1$	$ax + by + c = 0$	●○○○
5	The slope of the line $6x - 5y + 15 = 0$ is:	6/5	-6/5	5/6	-5/6	●○○○
6	Equation of a line parallel to the y -axis through (3, 5) is:	$y = 5$	$x = 3$	$y = 3$	$x = 5$	○○●○○
7	The x -intercept of the line $x + y = 5$ is:	5	-5	1	-1	●○○○
8	For two lines with slopes m_1, m_2 : $\tan \theta =$	$(m_2 - m_1) / (1 + m_1 m_2)$	$(m_2 + m_1) / (1 - m_1 m_2)$	$m_1 m_2 / (1 + m_1 + m_2)$	$(m_2 - m_1) / (1 - m_1 m_2)$	●○○○
9	Three points A, B, C are collinear if:	Slope of AB $\neq$ slope of BC	Slope of AB = slope of BC	$AB + BC = AC$	$AB \perp BC$	○○●○○
10	In normal form $x \cos \alpha + y \sin \alpha = p$, p represents:	Slope of the line	The y -intercept	Length of perpendicular from origin	The x -intercept	○○○●
11	The line $y = 0$ represents:	y -axis	x -axis	a point	a plane	○○●○○
12	Point of intersection of $9x - 7y = 0$ and $8x - 11y = 0$ is:	(0, 0)	(0, 7)	(9, 0)	(7, 9)	●○○○

SECTION B – Short Questions (11 × 3 = 33 Marks)

Attempt ALL 11 questions. Choose either the main question **OR** its alternative.

Q.	Question	Marks	OR	OR Question	Marks
2(i)	Find gradient of line with inclination 120° . State whether positive or negative and why.	3	OR	Find inclination of a line whose slope is -1.732.	3
2(ii)	Find the gradient and inclination of the line joining A(2, 6) and B(5, 8).	3	OR	Find gradient and inclination of the line joining C(-2, 4) and D(1, -3).	3
(iii)	A(-2, 6), B(7, -3): find slope of line (a) parallel to AB, (b) perpendicular to AB.	3	OR	Find x if slope of line through A(3, x) and B(5, 8) is 4.	3
(iv)	Using slopes, prove X(0, -3), Y(4, 7) and Z(6, 12) are collinear.	3	OR	Find y if points P(4, y), Q(5, 2) and R(6, $2y+1$) are collinear.	3
(v)	Find equation of line with slope = 2 and y -intercept = -3. Write in general form.	3	OR	Find equation of line through (-5, 7) with slope 4 using point-slope form.	3
(vi)	Find equation of line through (2, 4) and (-1, -4) using two-point form.	3	OR	Find equation of line with x -intercept = -6 and y -intercept = 5 using two-intercept form.	3
vii)	Reduce $6x - 5y + 15 = 0$ to slope-intercept form. State slope and y -intercept.	3	OR	Reduce $4x - 3y + 9 = 0$ to two-intercept form. State x -intercept and y -intercept.	3